

Physical Chemistry Formula Revision

Basic Concepts

$$\text{Average atomic mass} = \frac{a_1 x_1 + a_2 x_2 + a_3 x_3 \dots}{100}$$

$a_1, a_2, a_3 \dots$ - atomic mass of isotopes

$x_1, x_2, x_3 \dots$ - percentage abundance

Mole concept

$$\text{no. of moles, } n = \frac{\text{given mass}}{\text{molar mass}} = \frac{W}{M}$$

$$n = \frac{\text{no. of particles}}{\text{Avogadro's No.}} = \frac{N}{6.022 \times 10^{23}}$$

$$n = \frac{\text{vol. of gas}}{\text{vol. of gas at STP}} = \frac{V}{22.4 \text{ l}}$$

Concentration Terms

$$\text{Percentage, \%} = \frac{W_b \text{ or } V_b}{W \text{ or } V} \times 100$$

$$\text{Molarity, } M = \frac{n_b}{V(l)} = \frac{W_b \times 1000}{M_b \times V(ml)}$$

$$\text{Normality, } N = \frac{n_{eq}}{V(l)} = \frac{W_b \times 1000}{E_b \times V(ml)} \rightarrow \frac{M_b}{n_f}$$

$$\text{parts per million, ppm} = \frac{W_b}{W} \times 10^6$$

$$\text{Molality, } m = \frac{n_b}{W_a (kg)} = \frac{W_b \times 1000}{M_b \times W_a (g)}$$

$$\text{Mole fraction, } X_a = \frac{n_b}{n_a + n_b + n_c \dots}$$

Stoichiometry



$$\frac{\text{moles of A}}{a} = \frac{\text{moles of B}}{b} = \frac{\text{moles of C}}{c}$$

Limiting Rg: given amt. $\rightarrow$ lowest stoichiometric coeff.

Atomic Structure

Planck's Quantum Theory

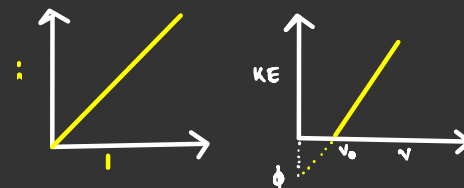
$$E = \frac{hc}{\lambda} \quad E (eV) = \frac{12400}{\lambda (A^\circ)}$$

Photo Electric Effect



$$E = \phi + KE_{max}$$

$$E = h\nu_0 + \frac{mv_{max}^2}{2}$$



De Broglie's Hypothesis

$$mc^2 = \frac{hc}{\lambda} \rightarrow \lambda = \frac{h}{mv} \quad \lambda = \frac{h}{\sqrt{2mKE}}$$

Bohr's Atomic model

$$mvr = \frac{nh}{2\pi} \quad r = 0.53 \times \frac{n^2}{Z} A^\circ$$

$$v = 2.18 \times 10^6 \times \frac{Z}{n} \text{ m/s}$$

$$E = -13.6 \times \frac{Z^2}{n^2} \text{ eV} \quad TE = \frac{pE}{2} = -KE$$

$$\frac{1}{\lambda} = R_H Z^2 \left[\frac{1}{n_1^2} - \frac{1}{n_2^2} \right] \quad R_H = 1.1 \times 10^7 \text{ m}^{-1}$$

max no. of spectral lines when e^- returns from n_2 to n_1 : $\left[\frac{n_2(n_2-1)}{2} \right]$

Heisenberg's Uncertainty Principle: $\Delta x \cdot \Delta p \gg \frac{h}{4\pi}$

Quantum Model

total nodes = $n-1$

angular nodes = l

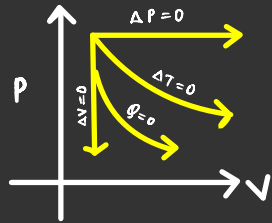
radial nodes = $n-1-l$

Angular momentum

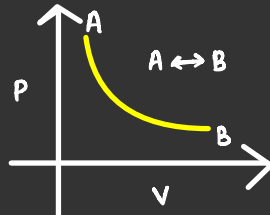
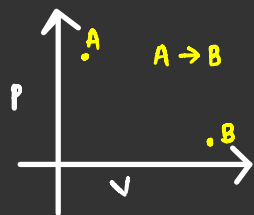
$$\text{orbital} = \sqrt{l(l+1)} \hbar \quad 2\pi$$

$$\text{spin} = \sqrt{s(s+1)} \hbar \quad 2\pi$$

Thermodynamic Process

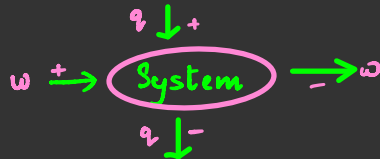


isothermal → $\Delta T = 0$
 isochoric → $\Delta V = 0$
 isobaric → $\Delta P = 0$
 adiabatic → $Q = 0$



1st Law of Thermodynamics

$$\Delta U = q + W$$



Heat Capacity

$$q \propto \Delta T$$

$$C = q / \Delta T \quad C_m = q / n \Delta T \quad C_s = q / m \Delta T$$

$$C_p: C_m \text{ at const. } P \quad C_v: C_m \text{ at const. } V$$

$$\Delta H = n C_p \Delta T \quad \Delta U = n C_v \Delta T$$

$$\text{Note: } C_p - C_v = R$$

$$\text{Poisson's ratio } \gamma : C_p / C_v$$

For isothermal $C = \infty$
 For adiabatic $C = 0$

	C_v	C_p	γ
monoatomic	$3R/2$	$5R/2$	$5/3$
di/triatomic (linear)	$5R/2$	$7R/2$	$7/5$
triatomic (non-linear)	$3R$	$4R$	$4/3$

Work Done

$$dW = -P_e \cdot dV$$

$$\text{For isobaric: } W = -nRT \Delta T$$

$$\text{For isochoric: } W = 0$$

For isothermal:

$$W_{rev} = 2.3nRT \log(V_2/V_1)$$

$$= 2.3nRT \log(P_2/P_1)$$

$$W_{irr} = -P_e (V_2 - V_1)$$

For adiabatic:

$$W_{rev} = \frac{(P_2 V_2 - P_1 V_1)}{\gamma - 1}$$

$$W_{irr} = \frac{(P_2 V_2 - P_1 V_1)}{\gamma - 1}$$

$$nC_v \Delta T = P \Delta V$$

For cyclic: $W = \text{area enclosed}$



Relation b/w ΔH and ΔU

$$\Delta U = q_p + W$$

$$\Delta U = q_p - P \Delta V$$

$$q_p = \Delta U + P \Delta V$$

$$\Delta H = \Delta U + P \Delta V$$

$$\Delta H = \Delta U + \Delta n_g RT$$

Entropy

$$\Delta S = q_{rev} / T$$

$$\Delta S = n C_v \ln(T_2/T_1) + n R \ln(V_2/V_1)$$

$$\Delta S = n C_p \ln(T_2/T_1) + n R \ln(P_1/P_2)$$

2nd Law of Thermodynamics

$$\Delta S_{univ} \geq 0$$

Gibbs free energy

$$G = H - TS$$

$$\Delta G = \Delta H - T \Delta S$$

$$\Delta G_{sys} = -T \Delta S_{univ}$$

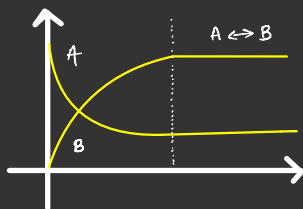
$\Delta G \rightarrow +ve$ non spont.
 $\Delta G \rightarrow -ve$ spont.
 $\Delta G \rightarrow 0$ equilibrium

$$\Delta G = \Delta G^\circ + 2.303 RT \log Q$$

at eq. $\Delta G = 0$; $Q = K_{eq}$

$$\Delta G^\circ = -2.303 RT \log K_{eq}$$

Equilibrium



Law of Mass Action



at eq. $R_f = R_b$

$$K_c = \frac{[C]^c [D]^d}{[A]^a [B]^b}$$

$$K_p = \frac{P_C^c P_D^d}{P_A^a P_B^b}$$

$$K_p = K_c (RT)^{\Delta n_g}$$

$K \uparrow \rightarrow$ Product stable

$K \downarrow \rightarrow$ reactant stable

Vant Hoff Equation

$$\log \frac{K_2}{K_1} = \frac{\Delta H}{2.303 R} \left[\frac{1}{T_1} - \frac{1}{T_2} \right]$$

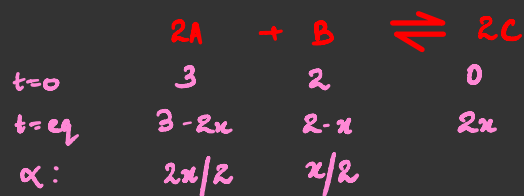
note:

For exo: $\Delta H = -ve \rightarrow T \uparrow \rightarrow K \downarrow$

For endo: $\Delta H = +ve \rightarrow T \uparrow \rightarrow K \uparrow$

Degree of Dissociation

$$\alpha = \frac{\text{no. of moles diss.}}{\text{initial no. of moles}}$$



$$K_c = \frac{\left(\frac{2x}{V}\right)^2}{\left(\frac{3-2x}{V}\right)^2 \left(\frac{2-x}{V}\right)}$$

note: $\alpha = \frac{D-d}{d(n-1)}$

$D = \sum VD$ of reactants = Molecular wt/2

$d = \sum VD$ of the reaction

$n =$ no. of moles of product for unit mole of reactants

Ionic Equilibrium

pH Calculation:

$$pH = -\log [H^+] \quad pOH = -\log [OH^-]$$

$$pH + pOH = 14 \quad (\text{at } 25^\circ C)$$

a) strong acid

$[H^+] > 10^{-6} M$ neglect H^+ from water
 $[H^+] < 10^{-6} M$ Consider H^+ from water

b) strong base similar method

$$[H^+] [OH^-] = 10^{-14}$$

c) mix of two SA and SB

$$[H^+] \text{ or } [OH^-] = \frac{|N_1 V_1 \pm N_2 V_2|}{V_1 + V_2}$$

$$pH + pOH = 14 \quad (\text{at } 25^\circ C)$$

d) weak acid and base

$$pH = \frac{1}{2} [pK_a - \log C] \quad \alpha = \sqrt{\frac{K_a}{C}}$$

$$[H^+] = \sqrt{K_a C}$$

e) mix of two WA or WB

$$[H^+] = \sqrt{K_{a1} C_1 + K_{a2} C_2}$$

$$[OH^-] = \sqrt{K_{b1} C_1 + K_{b2} C_2}$$

d) mix of WA and SA or WB and SB

$$[H^+] = \frac{-C_2 \pm \sqrt{C_2^2 + 4K_a C_1}}{2}$$

Salt Hydrolysis

a) WA + SB

$$K_H = \frac{K_w}{K_a}$$

$$\alpha = \sqrt{\frac{K_H}{C}}$$

$$pH = 7 + \frac{1}{2} [PK_a - \log C]$$

b) SA + WB

$$K_H = \frac{K_w}{K_b}$$

$$\alpha = \sqrt{\frac{K_H}{C}}$$

$$pH = 7 - \frac{1}{2} [PK_a - \log C]$$

c) WA + WB

$$K_H = \frac{K_w}{K_a K_b}$$

$$\alpha = \sqrt{K_H}$$

$$pH = 7 + \frac{1}{2} [PK_a - PK_b]$$

pH of Buffer Solution

acidic buffer

$$pH = PK_a + \log \frac{[salt]}{[acid]}$$

basic buffer

$$pH = PK_a + \log \frac{[salt]}{[base]}$$

Solutions

Henry's Law

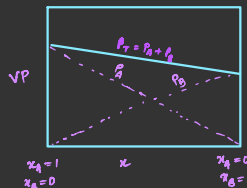
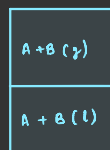
$$P = K_H x_B$$

Raoult's Law

$$P_T = P_A + P_B$$

$$P_T = P_A^0 x_A + P_B^0 x_B$$

$$P_T = (P_A^0 - P_B^0) x_A + P_B^0$$



In vapour phase

$$y_A = P_A / P_T \quad y_B = P_B / P_T$$

For non-volatile solute

$$P_T = P_A^0 x_A$$

Colligative Properties

Relative lowering in VP

$$P_T = P_A^0 x_A$$

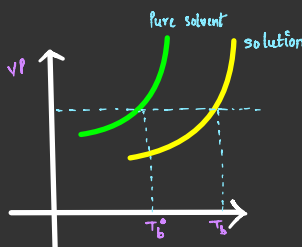
$$P_T = P_A^0 (1 - x_B)$$

$$x_B = \frac{P_A^0 - P_T}{P_A^0}$$

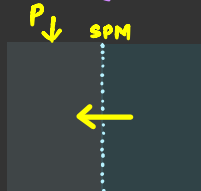
Elevation in B.P

$$\Delta T_b = K_b m$$

$$K_b = \frac{R T_b^{02} M}{1000 \Delta H_{vap}}$$



Osmotic Pressure



$$\pi = CRT$$

$P < \pi$ Osmosis

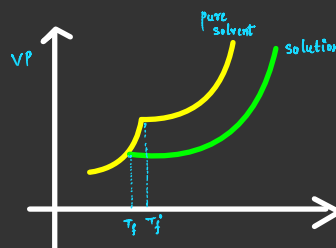
$P = \pi$ no flow

$P > \pi$ reverse osmosis

Depression in B.P

$$\Delta T_f = K_f m$$

$$K_f = \frac{R T_f^{02} M}{1000 \Delta H_{fus}}$$



Van't Hoff Factor

$$i = \frac{\text{normal molar mass}}{\text{abnormal molar mass}} = \frac{\text{observed CP}}{\text{calculated CP}}$$

dissociation $i = 1 + (n-1)\alpha$

$$i = \frac{\text{total no. of moles after dissociation}}{\text{total no. of moles before dissociation}} \quad \text{association}$$

association $i = 1 + (1/n-1)\alpha$

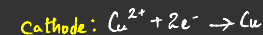
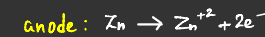
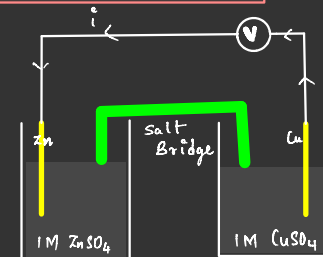
$$\frac{P_A^0 - P_T}{P_A^0} = i x_B$$

$$\Delta T_b = i K_b m$$

$$\Delta T_f = i K_f m$$

$$\pi = i CRT$$

Electrochemistry



$$E_{cell}^0 = E_C^0 - E_A^0$$

$$\Delta G^0 = -nFE_{cell}^0$$

$$\Delta G^0 = -RT \ln K_{eq}$$

Nernst Equation



$$E_{\text{cell}} = E_{\text{cell}}^{\circ} - \frac{0.059}{n} \log Q$$

$$E_{\text{cell}} = E_{\text{cell}}^{\circ} - \frac{0.059}{n} \log \frac{[C]^c}{[A]^a} \rho_B^b$$

at eq, $E_{\text{cell}} = 0$ and $Q = K_{\text{eq}}$

$$E_{\text{cell}}^{\circ} = \frac{0.059}{n} \log K_{\text{eq}}$$

Faraday's Law

1st law

$$W \propto Q$$

$$W = ZQ$$

$$W = \frac{E}{F} \cdot it$$

2nd law

$$W \propto E$$

$$\frac{W_1}{E_1} = \frac{W_2}{E_2}$$

Cell Characteristics

resistance, $R = \frac{\rho \cdot l}{A}$ ohm

resistivity, $\rho = \frac{R \cdot A}{l}$ ohm-cm

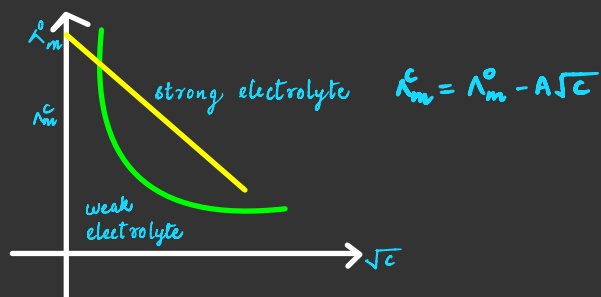
Conductance, $G = \frac{1}{R}$ ohm⁻¹ or S

conductivity, $K = \frac{1}{\rho}$ S cm⁻¹

$$\rightarrow K = \frac{1}{R \cdot A} = \frac{G^a}{R} \rightarrow \text{cell constant}$$

molar Conductivity, $\Lambda_m^c = \frac{K \times 1000}{M}$ S cm² mol⁻¹

Variation of Molar Conductivity



Kohlrausch Law



$$\Lambda_m^{\circ}(A_x B_y) = x \cdot \Lambda_m^{\circ}(A^{y+}) + y \cdot \Lambda_m^{\circ}(B^{x-})$$

$$\alpha = \frac{\Lambda_m^c}{\Lambda_m^{\circ}}$$

$$K_a = \frac{C\alpha^2}{1-\alpha}$$

$$\Lambda_m^{\circ}(\text{CH}_3\text{COOH}) = \Lambda_m^{\circ}(\text{CH}_3\text{COONa}) + \Lambda_m^{\circ}(\text{HCl}) - \Lambda_m^{\circ}(\text{NaCl})$$

Chemical kinetics

Rate of Reaction



$$\text{avg. rate} = \underbrace{-\frac{1}{a} \frac{\Delta[A]}{\Delta t}}_{\text{rate of disappearance}} = \underbrace{-\frac{1}{b} \frac{\Delta[B]}{\Delta t}}_{\text{rate of disappearance}} = \underbrace{+\frac{1}{c} \frac{\Delta[C]}{\Delta t}}_{\text{rate of appearance}} = \underbrace{+\frac{1}{d} \frac{\Delta[D]}{\Delta t}}_{\text{rate of appearance}}$$

Rate Law

$$\text{rate} = K [A]^m [B]^n \quad \text{unit of } K = (\text{conc})^{1-n} \text{ s}^{-1}$$

$$m = a$$

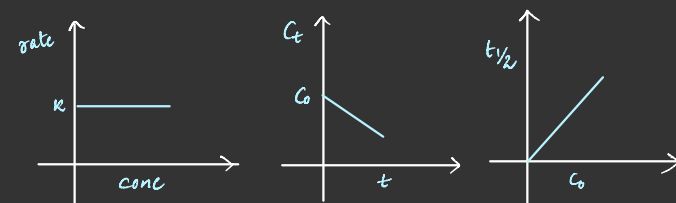
$$n = b$$

Integrated Rate Equations

Zero order Reactions

$$C_t = C_0 - kt$$

$$t_{1/2} = \frac{C_0}{2K}$$

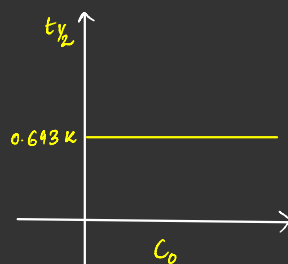
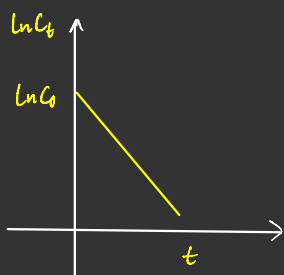
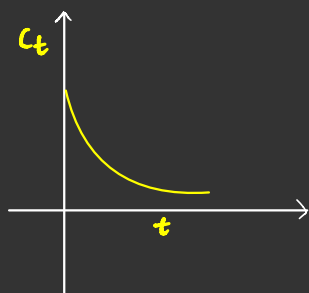
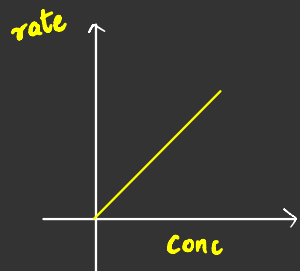


First order Reactions

$$C_t = C_0 e^{-kt}$$

$$k = \frac{2.303}{t} \log \frac{C_0}{C_t}$$

$$t_{1/2} = \frac{\ln 2}{k} = \frac{0.693}{k}$$



Integrated Rate Equations

Gas phase first order Reactions.



at $t=0$ P_0 0 0

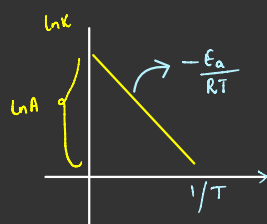
at $t=t$ $P_0 - P$ P P $\longrightarrow P_t = P_0 + P$
 $P = P_t - P_0$

$$k = \frac{2.303}{t} \log \frac{P_0}{P_0 - P}$$

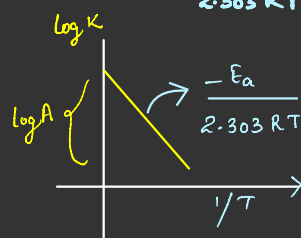
$$k = \frac{2.303}{t} \log \frac{P_0}{2P_0 - P_t}$$

Arrhenius Equation

$$\ln k = \ln A - \frac{E_a}{RT}$$



$$\log k = \ln A - \frac{E_a}{2.303 RT}$$



at temp T_1 , rate constant = k_1

at temp T_2 , rate constant = k_2

$$\log \frac{k_2}{k_1} = \frac{E_a}{2.303 R} \left[\frac{1}{T_1} - \frac{1}{T_2} \right]$$